

Vishal Kumar Rai

Ph.D. Candidate in Economics | Quantitative Macro & Life-Cycle Modeling

Indian Statistical Institute, Delhi

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Professional Summary

Ph.D. candidate in Economics at the Indian Statistical Institute, Delhi, specializing in structural and quantitative macroeconomics. My research develops dynamic life-cycle models of retirement, health, and labor supply under uncertainty, with a focus on environmental, epidemiological, and occupational risks. I combine continuous-time optimization and dynamic programming with microdata-based calibration (MEPS, CPS, and life tables) to quantify policy-relevant welfare and behavioral responses. Technical work is implemented in Julia and Python, with applications to public finance, social protection, and climate-health economics.

Skills

Programming: Julia, Python

Methods: Dynamic Programming, Structural Modeling, Calibration

Data: MEPS, CPS, CDC-NVSS life tables

Tools: L^AT_EX, Git

Education

Ph.D. Economics *2019–Present*
ISI Delhi
(Expected 2026)

M.A. Economics *2016–2019*
Delhi School of Economics

B.A. (Hons.) Business Economics *2012–2015*
University of Delhi

Languages

English (Fluent)
Hindi (Native)

Awards

JRF (ISI, 2019)
SRF (2021)
Top 25 DSE
CVS Rank 3

Teaching

TA, ISI Delhi (Prof. Chetan Ghate)
Macroeconomics I & Global Macroeconomics

- Led MSQE & PhD tutorials
- Assisted evaluation and coordination

Research Experience

Climate, Pollution and Endogenous Retirement

- Mechanism: environmental degradation affects mortality, productivity, and work disutility.
- Framework: stochastic continuous-time life-cycle model with endogenous retirement and health investment.
- Method: calibration to micro mortality and expenditure profiles; policy simulation.
- Contribution: retirement and willingness-to-pay for environmental quality in developing economies.

Endogenous Retirement under a Pandemic

- Mechanism: pandemic risk alters survival and labor-leisure trade-offs.
- Framework: life-cycle model with stochastic mortality under expected/unexpected shocks.
- Method: dynamic simulation with heterogeneous epidemiological scenarios.
- Contribution: breakdown of consumption smoothing and optimal policy responses.

Occupational Variation in Work Strain

- Mechanism: occupational heterogeneity in disutility and health risk.
- Framework: life-cycle model with occupation-specific work strain.
- Method: analytical + numerical simulation of heterogeneous exposure.
- Contribution: welfare and retirement distortions; policy implications.

Computational Tools

SwitchingControl.jl — Registered Julia package for endogenous regime-switching ODE systems using event-driven solvers, designed for macroeconomic and optimal control applications.